Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	Yaponomics
Selected Domain	Domain 2: Orbyte Smart SME Challenge
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth(Targets 8.2,8.3)
Project / Solution Name	Yapoz
Name (Leader)	Abdulrahman Imaad Mohamed
Email & Phone (Leader)	SWE2409099@xmu.edu.my
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

More than 97% of businesses are micro, small and medium enterprises (MSME) which account for 38% of GDP and 48% of employment (Securities Commission Malaysia, 2023). According to the Department of Statistics Malaysia (MyCensus 2020), approximately 13.8 million Malaysians are using WhatsApp. A bakery, salon or small shop's food stall is its shop counter.

The owner also serves as cook, cashier and accountant. Any incoming chat order is an order that has to be read, the price checked, stock checked, send bank info, wait for the transfer screen shot, and have to manually verify the screenshot. When owner is busy, replies are slow; prices are quoted inconsistently and same question is repeatedly answered. If the photo of the payment is not clear or edited, it is difficult to judge in the middle of a hurry and it is easy to make mistakes which will end up lost sale or unpaid order.

Such companies do not have the budget for custom software or employee training. General chatbots can have the ability to create prices and policies, and therefore owners cannot have confidence in their money.

This impacts: Owners and small teams that run day to day operations on WhatsApp and Telegram, and the customers who wait for answers. Why it matters: Hours of repetitive chat work are hours not spent growing the business, which slows down productivity (SDG 8.2) and the growth of micro and small enterprises (SDG 8.3). The difficulty lies in eliminating the repetitive work in a way that does not eliminate the owner's control.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Yapoz is an Artificial Intelligence (AI) front desk for small businesses that are already selling on WhatsApp and Telegram. It makes repetitive chat work, and leaves all decisions that relate to it to the owner.

One complete journey.

(1) Trigger: Customer sends a message to the business.

(2) Connected data: Only the owner can enter what is shown, e.g. menu, size, price, delivery areas, policies and payment accounts, etc.

(3) Decision logic: AI agent responds using tools to look up these facts; it cannot create a price/policy. Order price is by code and not by model number, and is checked against inventory. The customer is provided with information on how to pay and requested for a receipt. The vision model reads the photo of the receipt, and the plain rules compare it to the order's details: amount, payee, date, status and reuse of the receipt in another order.

(4) Human approval: order is pending. The owner views order, receipt, and each check and confirms or cancels. When there is no receipt confirming is locked, and the AI never marks paid items.

(5) Visible outcome: The customer is informed in chat, stock is updated and each step is logged when in an activity log.

Any questions that the AI is not able to answer are added to a Gaps list. The owner responds and the response is taken from the next message, therefore there are no repeated explanations from the same person.

The roles (owner and service desk) and autopilot/copilot mode per channel keep people in control.

Technologies: GenAI APIs (Groq-hosted open models and Google Gemini for vision), PostgreSQL database (Supabase), cloud hosting (Vercel). Every business is a workspace and one product can serve a number of SMEs and can be connected to Orbyte as a re-using component.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

This application was built using Next.js 16 (React, TypeScript) on Vercel serverless functions. The owner dashboard (Inbox, Orders, Knowledge, Settings) and webhook APIs are provided using one codebase.

Database: PostgreSQL on Supabase + Drizzle ORM. Each record is related to a business workspace (row-level security policies), each jobs table functions as a retrying queue and each listen/notify event pushes real-time updates to the dashboard.

Channels: include WhatsApp Cloud API (Meta), Telegram Bot API (grammY) and Instagram messages. Webhooks from Meta or Telegram are checked (with HMAC signature for Meta and with secret token for Telegram).

AI: a provider router that automatically switches between Groq's (OpenAI gpt-oss) models and Google Gemmas. A tool-calling agent responds based on what it knows from the owner's knowledge base. The payment receipts are read by the Gemini vision, and a deterministic rule engine compares the read receipt with the order.

Security and quality: AES-256-GCM encrypted credentials, bcrypt passwords, signed sessions, role-based access, and 110 automated tests run against a throwaway database in which the AI is stubbed.

AI coding tools: Claude Code, Replit Agent (initial project structure).

Sponsor platforms: Yes. For the initial project structure, Replit Agent, the official SDC platform was used. Unfortunately, its credits were up and development was continued with Claude Code, and the team still has screenshots of the emptied credits as proof. The team didn't choose RunPod since they do not train a model nor host one. The public APIs of Groq and Google Gemini allow access to the models. It is deployed on Vercel at yapoz.vercel.app.

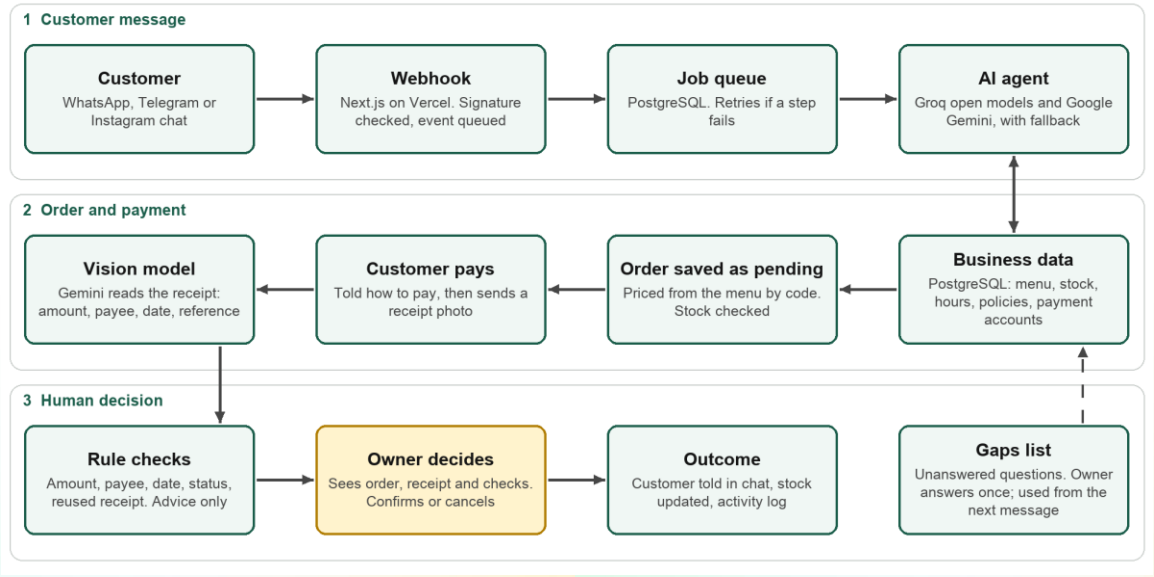


Figure 1. System block diagram: from a customer message to the owner's decision.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise. We were customers of a real business on WhatsApp and Telegram, and wrote just like real customers, in the slang and using vague messages such as "can i order", changing minds and adding photos. We saw where the owner had to get in. **Define.** We identified three common pain points: pricing of orders that are determined by whom the answer comes from, payment receipts that can't be easily checked and the same unanswered questions showing up again. **Ideate.** The AI didn't take into account items which it may not know, such as the prices from the menu and answers from the owner's knowledge, and the owner's final say. **Prototype and test.** Every pain was now a feature for short loops: orders on the menu, reading the receipt with advisory checks, and the Gaps list. Any failure that was discovered this way, e.g. the AI fails during the middle of an order, was addressed and was covered by automated tests (110 today). We have not yet tested with real owners of SMEs, that is our first step in Phase 2. **Deployment.** The app is deployed on Vercel and Supabase, and includes a health endpoint, repeatable database migrations, and a synthetic demo business, which allows judges to try it without any real customer data.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 8, Target 8.2:** "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation." **Target 8.3:** "...encourage the formalization and growth of micro-, small- and medium-sized enterprises." Yapoz raises productivity by automating repetitive chat order and payment work, and supports micro business growth with a tool that needs no training.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Handling one chat order by hand takes **5 manual steps**: read the message, look up the price, check stock, send payment details, check the payment receipt. Today 100% of chat orders are handled this way and 0% arrive pre-priced or pre-checked. Scale: MSMEs are more than 97% of Malaysian businesses [1] and about 13.8 million Malaysians use WhatsApp [2]. No public Malaysian figure exists for minutes per order, so we will time the manual process on 20 synthetic orders at the start of Phase 2 and report it.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Manual steps the owner performs per chat order** (baseline: 5). Supporting measures: the share of orders that reach the owner already priced and stock-checked, and the share of receipts checked by the system before the owner sees them.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Cut manual steps per order from 5 to 1 (the owner's approval): an **80% reduction**. At least **90%** of test orders arrive priced and stock-checked, and **100%** of uploaded receipts are checked for amount, payee, date and reuse, so the owner only looks closely at the flagged ones.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [ ] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js, React, React DOM, SWR, Zod, jose, clsx, date-fns, qrcode, unpdf, pdf-lib, Tailwind CSS, tsx, ESLint	code (libraries)	MIT
Radix UI (dialog, dropdown, popover, switch, tabs, tooltip), Motion, Lenis	code (UI libraries)	MIT
Three.js, React Three Fiber, drei, postprocessing	code (3D landing page)	MIT (postprocessing: Zlib)
GSAP and @gsap/react	code (animation)	GSAP Standard "no charge" license
Drizzle ORM, OpenAI SDK (used for OpenAI-compatible APIs), TypeScript, Playwright	code	Apache-2.0
postgres (postgres.js driver)	code	Unlicense
bcryptjs	code	BSD-3-Clause

Bricolage Grotesque, JetBrains Mono, Instrument Serif (via Fontsource)	fonts	OFL-1.1
OpenAI gpt-oss-120b and gpt-oss-20b, run by Groq	pre-trained model (API)	Apache-2.0
Qwen model served by Groq (fallback only)	pre-trained model (API)	As published on Groq's model page
Google Gemini flash models (receipt reading, chat fallback)	pre-trained model (API)	Google Gemini API terms
WhatsApp Cloud API (Meta), Telegram Bot API	external services (API)	Provider terms
Replit Agent and Claude Code (AI coding assistants used while building)	AI development tools	Provider terms
Demo data for Forno Sette and Kasih Hair Studio (menus, chats, receipts)	dataset (synthetic)	Created by the team
grammY (Telegram), Anthropic SDK, Drizzle Kit	grammY (Telegram), Anthropic SDK, Drizzle Kit	MIT

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]: Abdulrahman Imaad Mohamed

[Date] : 24/10/2026